# AI Assisted Coding

## Assignment – 11.4

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### Task 1: Implementing a Stack (LIFO)

- Task: Use AI to help implement a Stack class in Python with the following operations: push(), pop(), peek(), and is_empty().
- Instructions:
  - Ask AI to generate code skeleton with docstrings.
  - Test stack operations using sample data.
  - Request AI to suggest optimizations or alternative implementations (e.g., using collections.deque).
- Expected Output: A working Stack class with proper methods, Google-style docstrings, and inline comments for tricky parts.

Code :

```
# Create a new stack
my_stack = Stack()

# Check if the stack is empty
print(f"Is the stack empty? {my_stack.is_empty()}")

# Push some items onto the stack
my_stack.push(10)
my_stack.push(20)
my_stack.push(30)

# Check if the stack is empty after pushing
print(f"Is the stack empty? {my_stack.is_empty()}")

# Peek at the top item
print(f"Top item: {my_stack.peek()}")

# Pop items from the stack
print(f"Popped item: {my_stack.pop()}")
print(f"Popped item: {my_stack.pop()}")

# Peek at the top item after popping
print(f"Top item after popping: {my_stack.peek()}")

# Pop the last item
print(f"Popped item: {my_stack.pop()}")

# Check if the stack is empty again
print(f"Is the stack empty? {my_stack.is_empty()}")

# Try to pop from an empty stack (will raise an error)
try:
    my_stack.pop()
except IndexError as e:
    print(f"Error: {e}")
```

Output:

```
Is the stack empty? True
Is the stack empty? False
Top item: 30
Popped item: 30
Popped item: 20
Top item after popping: 10
Popped item: 10
Is the stack empty? True
Error: pop from empty stack
```

Explanation:

1. my_stack = Stack(): This line creates a new, empty instance of your Stack class and assigns it to the variable my_stack.

2. print(f"Is the stack empty? {my_stack.is_empty()}"): This calls the is_empty() method on the my_stack object to check if the stack is currently empty. Since it was just created, it will print True.
3. my_stack.push(10), my_stack.push(20), my_stack.push(30): These lines use the push() method to add the integers 10, 20, and 30 onto the top of the stack, in that order.
4. print(f"Is the stack empty? {my_stack.is_empty()}"): This checks if the stack is empty again after adding items. It will now print False.
5. print(f"Top item: {my_stack.peek()}"): This calls the peek() method, which returns the item at the top of the stack (which is 30) without removing it.
6. print(f"Popped item: {my_stack.pop()}"), print(f"Popped item: {my_stack.pop()}"): These lines call the pop() method twice. Each call removes and returns the item from the top of the stack. First 30 is popped, then 20.
7. print(f"Top item after popping: {my_stack.peek()}"): After the two pops, the item now at the top of the stack is 10. This line calls peek() again to show this.
8. print(f"Popped item: {my_stack.pop()}"): This pops the last remaining item (10) from the stack.
9. print(f"Is the stack empty? {my_stack.is_empty()}"): This checks the stack's empty status again. Since all items have been popped, it will print True.
10. try...except IndexError as e: print(f"Error: {e}"): This block attempts to call my_stack.pop() one more time. Since the stack is now empty, calling pop() will raise an IndexError. The try...except block catches this specific error, and the code inside the except block is executed, printing the error message "pop from empty stack". This demonstrates how the pop() method handles an empty stack.

## Task 2: Queue Implementation with Performance Review

• Task: Implement a Queue with enqueue(), dequeue(), and is_empty() methods.
• Instructions:
o First, implement using Python lists.
o Then, ask AI to review performance and suggest a more e icient implementation (using collections.deque).
• Expected Output: o Two versions of a queue: one with lists and one optimized with deque, plus an AI-generated performance comparison.

Code:

```python
# Code for ListQueue implementation
class ListQueue:
    """
    A simple Queue implementation using a Python list.

    A queue is a linear data structure that follows the FIFO (First-In, First-Out) principle.
    """

    def __init__(self):
        """
        Initializes an empty Queue.
        """
        self._items = []

    def enqueue(self, item):
        """
        Adds an item to the rear of the queue.

        Args:
            item: The item to be added to the queue.
        """
        self._items.append(item)

    def dequeue(self):
        """
        Removes and returns the item from the front of the queue.

        Returns:
            The item at the front of the queue.

        Raises:
            IndexError: If the queue is empty.
        """
        if not self.is_empty():
            return self._items.pop(0) # Removing from the front of a list is O(n)
```

```
            else:
                raise IndexError("dequeue from empty queue")

        def is_empty(self):
            """
            Checks if the queue is empty.

            Returns:
                True if the queue is empty, False otherwise.
            """
            return len(self._items) == 0

        def peek(self):
            """
            Returns the item at the front of the queue without removing it.

            Returns:
                The item at the front of the queue.

            Raises:
                IndexError: If the queue is empty.
            """
            if not self.is_empty():
                return self._items[0]
            else:
                raise IndexError("peek from empty queue")
```

```python
# Test the list-based Queue implementation
print("--- Testing ListQueue ---")
my_list_queue = ListQueue()
print(f"List Queue - Is empty? {my_list_queue.is_empty()}")
my_list_queue.enqueue(10)
my_list_queue.enqueue(20)
my_list_queue.enqueue(30)
print(f"List Queue - Is empty? {my_list_queue.is_empty()}")
print(f"List Queue - Front item: {my_list_queue.peek()}")
print(f"List Queue - Dequeued item: {my_list_queue.dequeue()}")
print(f"List Queue - Dequeued item: {my_list_queue.dequeue()}")
print(f"List Queue - Front item after dequeuing: {my_list_queue.peek()}")
print(f"List Queue - Dequeued item: {my_list_queue.dequeue()}")
print(f"List Queue - Is empty? {my_list_queue.is_empty()}")
try:
    my_list_queue.dequeue()
except IndexError as e:
    print(f"List Queue - Error: {e}")
print("-" * 25)


# Code for DequeQueue implementation
from collections import deque

class DequeQueue:
    """
    A Queue implementation using collections.deque.
    """
    def __init__(self):
        """
        Initializes an empty DequeQueue.
        """
        self._items = deque()

    def enqueue(self, item):
        """
        Adds an item to the rear of the queue.
```

```
        Args:
            item: The item to be added to the queue.
        """
        self._items.append(item) # append to the right is O(1)

    def dequeue(self):
        """
        Removes and returns the item from the front of the queue.

        Returns:
            The item at the front of the queue.

        Raises:
            IndexError: If the queue is empty.
        """
        if not self.is_empty():
            return self._items.popleft() # pop from the left is O(1)
        else:
            raise IndexError("dequeue from empty queue")

    def is_empty(self):
        """
        Checks if the queue is empty.

        Returns:
            True if the queue is empty, False otherwise.
        """
        return len(self._items) == 0

    def peek(self):
        """
        Returns the item at the front of the queue without removing it.

        Returns:
            The item at the front of the queue.
```

```
        Returns:
            The item at the front of the queue.

        Raises:
            IndexError: If the queue is empty.
        """
        if not self.is_empty():
            return self._items[0]
        else:
            raise IndexError("peek from empty queue")

# Test the collections.deque-based Queue implementation
print("--- Testing DequeQueue ---")
my_deque_queue = DequeQueue()
print(f"Deque Queue - Is empty? {my_deque_queue.is_empty()}")
my_deque_queue.enqueue(100)
my_deque_queue.enqueue(200)
my_deque_queue.enqueue(300)
print(f"Deque Queue - Is empty? {my_deque_queue.is_empty()}")
print(f"Deque Queue - Front item: {my_deque_queue.peek()}")
print(f"Deque Queue - Dequeued item: {my_deque_queue.dequeue()}")
print(f"Deque Queue - Dequeued item: {my_deque_queue.dequeue()}")
print(f"Deque Queue - Front item after dequeuing: {my_deque_queue.peek()}")
print(f"Deque Queue - Dequeued item: {my_deque_queue.dequeue()}")
print(f"Deque Queue - Is empty? {my_deque_queue.is_empty()}")
try:
    my_deque_queue.dequeue()
except IndexError as e:
    print(f"Deque Queue - Error: {e}")
print("-" * 25)
```

Output :

```
--- Testing ListQueue ---
List Queue - Is empty? True
List Queue - Is empty? False
List Queue - Front item: 10
List Queue - Dequeued item: 10
List Queue - Dequeued item: 20
List Queue - Front item after dequeuing: 30
List Queue - Dequeued item: 30
List Queue - Is empty? True
List Queue - Error: dequeue from empty queue
-------------------------
--- Testing DequeQueue ---
Deque Queue - Is empty? True
Deque Queue - Is empty? False
Deque Queue - Front item: 100
Deque Queue - Dequeued item: 100
Deque Queue - Dequeued item: 200
Deque Queue - Front item after dequeuing: 300
Deque Queue - Dequeued item: 300
Deque Queue - Is empty? True
Deque Queue - Error: dequeue from empty queue
-------------------------
```

## Task 3: Singly Linked List with Traversal

- Task: Implement a Singly Linked List with operations: insert_at_end(), delete_value(), and traverse().
- Instructions:
  - o Start with a simple class-based implementation (Node, LinkedList).
  - o Use AI to generate inline comments explaining pointer updates (which are non-trivial). o Ask AI to suggest test cases to validate all operations.
- Expected Output: o A functional linked list implementation with clear comments explaining the logic of insertions and deletions Code:

```
# Create an instance of the LinkedList class
my_list = LinkedList()

# Insert several elements into the list
print("Inserting elements:")
my_list.insert_at_end(10)
my_list.insert_at_end(20)
my_list.insert_at_end(30)
my_list.insert_at_end(40)

# Traverse and print the list to show the initial state
print("Initial list:")
my_list.traverse()

# Delete a value from the middle of the list
print("\nDeleting value 30:")
my_list.delete_value(30)

# Traverse and print the list again to show the effect of the deletion
print("List after deleting 30:")
my_list.traverse()

# Delete the head of the list
print("\nDeleting head (value 10):")
my_list.delete_value(10)

# Traverse and print the list to show the updated list
print("List after deleting 10:")
my_list.traverse()

# Attempt to delete a value that does not exist in the list
print("\nAttempting to delete non-existent value 50:")
my_list.delete_value(50)

# Traverse and print the list to confirm no changes were made
print("List after attempting to delete 50:")
my_list.traverse()
```

```
# Traverse and print the list to confirm no changes were made
print("List after attempting to delete 50:")
my_list.traverse()

# Insert another element and then delete the only remaining element in the list
print("\nInserting 50 and deleting 20 and 40 and 50:")
my_list.insert_at_end(50)
my_list.delete_value(20)
my_list.delete_value(40)
my_list.delete_value(50)


# Traverse and print the empty list
print("List after deleting all elements:")
my_list.traverse()

# Attempt to delete from an empty list
print("\nAttempting to delete from an empty list:")
my_list.delete_value(100)
```

Output:

```
Inserting elements:
Initial list:
10 -> 20 -> 30 -> 40 -> None

Deleting value 30:
List after deleting 30:
10 -> 20 -> 40 -> None

Deleting head (value 10):
List after deleting 10:
20 -> 40 -> None

Attempting to delete non-existent value 50:
Value 50 not found in the list.
List after attempting to delete 50:
20 -> 40 -> None

Inserting 50 and deleting 20 and 40 and 50:
List after deleting all elements:
None

Attempting to delete from an empty list:
List is empty. Cannot delete.
```

Explanation :

1. my_list = LinkedList(): This line creates a new, empty instance of your LinkedList class.
2. print("Inserting elements:"): This simply prints a header to indicate the start of insertion tests.
3. my_list.insert_at_end(10), my_list.insert_at_end(20), my_list.insert_at_end(30), my_list.insert_at_end(40): These lines call the insert_at_end() method to add the values 10, 20, 30, and 40 to the end of the linked list. The list will become 10 -> 20 -> 30 -> 40 -> None.
4. print("Initial list:"): Prints a header.
5. my_list.traverse(): Calls the traverse() method to print the current elements of the list. This confirms the initial insertions.
6. print("\nDeleting value 30:"): Prints a header for the deletion test.
7. my_list.delete_value(30): Calls delete_value() to remove the node with the value 30 from the list. The list should become 10 -> 20 -> 40 -> None.
8. print("List after deleting 30:"): Prints a header.
9. my_list.traverse(): Calls traverse() to show the list after deleting 30.
10. print("\nDeleting head (value 10):"): Prints a header for deleting the head node.
11. my_list.delete_value(10): Calls delete_value() to remove the node with the value 10, which is currently the head. The list should become 20 -> 40 -> None.
12. print("List after deleting 10:"): Prints a header.
13. my_list.traverse(): Calls traverse() to show the list after deleting the head.
14. print("\nAttempting to delete non-existent value 50:"): Prints a header for testing deletion of a non-existent value.
15. my_list.delete_value(50): Calls delete_value() to try and remove 50, which is not in the list. The delete_value method should handle this gracefully (e.g., by printing a message) and the list should remain unchanged: 20 -> 40 -> None.
16. print("List after attempting to delete 50:"): Prints a header.
17. my_list.traverse(): Calls traverse() to confirm the list is unchanged.
18. print("\nInserting 50 and deleting 20 and 40 and 50:"): Prints a header for testing deletion of multiple elements, including inserting a new one first.

19. my_list.insert_at_end(50): Inserts 50 at the end: 20 -> 40 -> 50 -> None.
20. my_list.delete_value(20), my_list.delete_value(40), my_list.delete_value(50): These lines sequentially delete the remaining elements (20, then 40, then 50) until the list is empty.
21. print("List after deleting all elements:"): Prints a header.
22. my_list.traverse(): Calls traverse() on the now empty list, which should just print None.
23. print("\nAttempting to delete from an empty list:"): Prints a header for testing deletion from an empty list.
24. my_list.delete_value(100): Calls delete_value() on the empty list. The method should handle this by printing a message like "List is empty. Cannot delete."

## Task 4: Binary Search Tree (BST)

- Task: Implement a Binary Search Tree with methods for insert(), search(), and inorder_traversal().
- Instructions: o Provide AI with a partially written Node and BST class.

o Ask AI to complete missing methods and add docstrings.
o Test with a list of integers and compare outputs of search() for present vs absent elements.
• Expected Output:
o A BST class with clean implementation, meaningful docstrings, and correct traversal output.

Code:

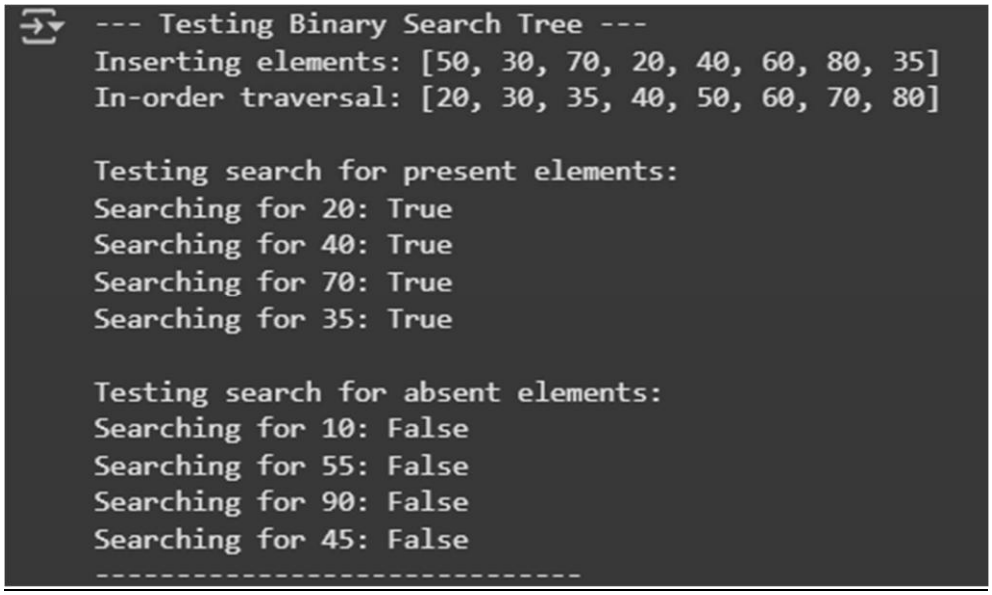
```
class Node:
    """
    Represents a node in a Binary Search Tree.
    """
    def __init__(self, key):
        """
        Initializes a new Node with a given key.

        Args:
            key: The value to be stored in the node.
        """
        self.key = key
        self.left = None
        self.right = None

class BST:
    """
    Represents a Binary Search Tree.
    """
    def __init__(self):
        """
        Initializes an empty Binary Search Tree.
        """
        self.root = None

    def insert(self, key):
        """
        Inserts a new node with the given key into the BST.

        Args:
            key: The value to be inserted.
        """
        if self.root is None:
            self.root = Node(key)
        else:
            self._insert_recursive(self.root, key)
```

```
    def _insert_recursive(self, node, key):
        """
        Helper function for recursive insertion.

        Args:
            node: The current node being considered.
            key: The value to be inserted.
        """
        if key < node.key:
            if node.left is None:
                node.left = Node(key)
            else:
                self._insert_recursive(node.left, key)
        else: # key >= node.key (allowing duplicates to the right)
            if node.right is None:
                node.right = Node(key)
            else:
                self._insert_recursive(node.right, key)

    def search(self, key):
        """
        Searches for a node with the given key in the BST.

        Args:
            key: The value to search for.

        Returns:
            True if the key is found, False otherwise.
        """
        return self._search_recursive(self.root, key)

    def _search_recursive(self, node, key):
        """
        Helper function for recursive search.
```

```
        Args:
            node: The current node being considered.
            key: The value to search for.

        Returns:
            True if the key is found, False otherwise.
        """
        if node is None:
            return False
        if node.key == key:
            return True
        elif key < node.key:
            return self._search_recursive(node.left, key)
        else:
            return self._search_recursive(node.right, key)

    def inorder_traversal(self):
        """
        Performs an in-order traversal of the BST and returns a list of keys.

        Returns:
            A list containing the keys in ascending order.
        """
        result = []
        self._inorder_recursive(self.root, result)
        return result

    def _inorder_recursive(self, node, result):
        """
        Helper function for recursive in-order traversal.

        Args:
            node: The current node being considered.
            result: The list to store the traversal results.
        """
        if node:
            self._inorder_recursive(node.left, result)
            result.append(node.key)
```

```
            if node:
                self._inorder_recursive(node.left, result)
                result.append(node.key)
                self._inorder_recursive(node.right, result)

    # Test the BST implementation
    print("--- Testing Binary Search Tree ---")

    # Create a new BST
    bst = BST()

    # List of integers to insert
    elements_to_insert = [50, 30, 70, 20, 40, 60, 80, 35]

    # Insert elements into the BST
    print("Inserting elements:", elements_to_insert)
    for element in elements_to_insert:
        bst.insert(element)

    # Perform in-order traversal
    print("In-order traversal:", bst.inorder_traversal())

    # Test search for present elements
    print("\nTesting search for present elements:")
    present_elements = [20, 40, 70, 35]
    for element in present_elements:
        found = bst.search(element)
        print(f"Searching for {element}: {found}")

    # Test search for absent elements
    print("\nTesting search for absent elements:")
    absent_elements = [10, 55, 90, 45]
    for element in absent_elements:
        found = bst.search(element)
        print(f"Searching for {element}: {found}")

    print("-" * 30)
```

Output:




## Task 5: Graph Representation and BFS/DFS Traversal

- Task: Implement a Graph using an adjacency list, with traversal methods BFS() and DFS().
- Instructions:

o Start with an adjacency list dictionary. o Ask AI to generate BFS and DFS implementations with inline

comments. o Compare recursive vs iterative DFS if suggested by AI.

- Expected Output: o A graph implementation with BFS and DFS traversal methods, with AI-generated comments explaining traversal steps.

Code:

```python
from collections import deque

class Graph:
    """
    Represents a graph using an adjacency list.
    """
    def __init__(self):
        # Initialize an empty dictionary to store the adjacency list
        self.adj_list = {}

    def add_edge(self, u, v):
        """
        Adds an edge between nodes u and v (for an undirected graph).

        Args:
            u: The first node.
            v: The second node.
        """
        # Add edge from u to v
        if u not in self.adj_list:
            self.adj_list[u] = []
        self.adj_list[u].append(v)

        # Add edge from v to u (for undirected graph)
        if v not in self.adj_list:
            self.adj_list[v] = []
        self.adj_list[v].append(u)

    def bfs(self, start_node):
        """
        Performs a Breadth-First Search (BFS) traversal starting from a given node.

        Args:
            start_node: The node to start the traversal from.
        """
        print(f"BFS traversal starting from node {start_node}:", end=" ")
        # Initialize a queue for BFS and add the start node
        queue = deque([start_node])
```

```
        # Continue as long as the queue is not empty
        while queue:
            # Dequeue a node from the front of the queue
            current_node = queue.popleft()
            # Process the current node (e.g., print it)
            print(current_node, end=" ")

            # Iterate through the neighbors of the current node
            # Use .get() with a default empty list to handle nodes with no neighbors
            for neighbor in self.adj_list.get(current_node, []):
                # If the neighbor has not been visited
                if neighbor not in visited:
                    # Mark the neighbor as visited
                    visited.add(neighbor)
                    # Enqueue the neighbor
                    queue.append(neighbor)
        print() # Print a newline at the end of traversal

    def dfs_iterative(self, start_node):
        """
        Performs an iterative Depth-First Search (DFS) traversal starting from a given node using a stack.

        Args:
            start_node: The node to start the traversal from.
        """
        print(f"DFS iterative traversal starting from node {start_node}:", end=" ")
        # Initialize a stack for DFS and add the start node
        stack = [start_node]
        # Initialize a set to keep track of visited nodes
        visited = set()

        # Continue as long as the stack is not empty
        while stack:
            # Pop a node from the top of the stack
            current_node = stack.pop()

            # If the current node has not been visited
            if current_node not in visited:
```

```
                # Mark the current node as visited
                visited.add(current_node)

                # Iterate through the neighbors of the current node
                # Reverse the neighbors to maintain a consistent exploration order (e.g., smaller nodes first if adjacer
                for neighbor in reversed(self.adj_list.get(current_node, [])):
                    # If the neighbor has not been visited
                    if neighbor not in visited:
                        # Push the neighbor onto the stack
                        stack.append(neighbor)
        print() # Print a newline at the end of traversal


    def _dfs_recursive_helper(self, node, visited):
        """
        Helper function for recursive Depth-First Search.

        Args:
            node: The current node being considered.
            visited: A set to keep track of visited nodes.
        """
        # Mark the current node as visited
        visited.add(node)
        # Process the current node (e.g., print it)
        print(node, end=" ")

        # Iterate through the neighbors of the current node
        for neighbor in self.adj_list.get(node, []):
            # If the neighbor has not been visited
            if neighbor not in visited:
                # Recursively call the helper function for the neighbor
                self._dfs_recursive_helper(neighbor, visited)


    def dfs_recursive(self, start_node):
        """
        Performs a recursive Depth-First Search (DFS) traversal starting from a given node.
```

```
        print(f"DFS recursive traversal starting from node {start_node}:", end=" ")
        # Initialize an empty set to keep track of visited nodes
        visited = set()
        # Call the recursive helper function starting from the start node
        self._dfs_recursive_helper(start_node, visited)
        print() # Print a newline at the end of traversal


# Example usage and testing:

# Instantiate the Graph class
g = Graph()

# Add edges to create a sample graph
g.add_edge('A', 'B')
g.add_edge('A', 'C')
g.add_edge('B', 'D')
g.add_edge('C', 'E')
g.add_edge('D', 'E')
g.add_edge('D', 'F')
g.add_edge('E', 'F')

# Display the adjacency list (optional, for verification)
# print("Adjacency List:", g.adj_list)

# Define the starting node for traversals
start_node = 'A'

# Perform BFS traversal
g.bfs(start_node)

# Perform iterative DFS traversal
g.dfs_iterative(start_node)

# Perform recursive DFS traversal
g.dfs_recursive(start_node)
```

Output:

```
BFS traversal starting from node A: A B C D E F
DFS iterative traversal starting from node A: A B D E C F
DFS recursive traversal starting from node A: A B D E C F
```